9. Partial Identification

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Partial Identification* (Prof. Sanghack Lee, GSDS SNU)

The central question is what can still be said about $P(y \mid \mathrm{do}(x))$ when it is not identifiable: instead of a point, we seek the tightest interval $[a(y;x), b(y;x)] \subseteq [0,1]$ consistent with the data and the graph.

These notes consolidate the slides, prove the natural bounds and their tightness, construct the canonical (response-type) model for the instrumental-variable setting, write the resulting linear program in full and verify one of its closed-form bounds, and present bounds with partially observed confounders. Physical PDF page numbers are listed in Section 5.

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1 The partial identification problem

Purpose. Extend the yes/no identification question to bounds.

Recall that $P(y \mid \mathrm{do}(x))$ is identifiable if it is uniquely determined by $P(\mathbf{v})$ and $\mathcal{G}$. When it is not, an inference engine given the query, the graph and data such as $P(x, y, z)$ answers "no". We then ask whether a non-trivial bound can be given:

$$P(y \mid \mathrm{do}(x)) \in [a(y; x),\, b(y; x)] \subseteq [0, 1].$$

Definition 1.1 (Partial identifiability)**.** The effect of X on Y is partially identifiable given $\mathcal{G}$ if $P(\mathbf{V})$ determines a bound $P(y \mid \mathrm{do}(x)) \in [a, b]$ that is a proper subset of $[0, 1]$.

Identifiable $\Rightarrow$ partially identifiable (with $a = b$, ignoring the degenerate case), but not conversely.

2 Natural bounds

Purpose. Derive the simplest bounds, valid for any confounded pair, and show they cannot be improved without more structure.

Theorem 2.1 (Natural bounds; Manski 1990, Robins 1989)**.** *For an SCM compatible with $X \to Y$, $X \leftrightarrow Y$ (latent U), given $P(x, y)$,*

$$a(y; x) = P(y, x) \;\le\; P(y \mid \mathrm{do}(x)) \;\le\; P(y, x) + 1 - P(x) = b(y; x).$$

Proof. With $X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x, u)$, the truncated product gives

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_u P(y \mid x, u) P(u) = \sum_u P(y \mid x, u)\big(P(u, x) + P(u) - P(u, x)\big) \\ &= \sum_u P(y, x, u) + \sum_u P(y \mid x, u)\big(P(u) - P(u, x)\big) \\ &= P(y, x) + \sum_u P(y \mid x, u)\big(P(u) - P(u, x)\big). \end{aligned}$$

Since $0 \le P(y \mid x, u) \le 1$ and $P(u) - P(u, x) \ge 0$, the last sum lies between 0 and $\sum_u (P(u) - P(u, x)) = 1 - P(x)$. □

The width of the interval is $1 - P(x)$: the probability mass of units that did not receive x, about whom the data say nothing under $\mathrm{do}(x)$. Equivalently $b(y; x) = 1 - P(x, y')$ with y' the complement of y.

Example 2.2. $X, Y, U \in \{0, 1\}$, $X \leftarrow U$, $Y \leftarrow X \oplus U$, $P(U = 0) = 0.9$. Observationally $X = U$ and $Y = 0$ always: $P(X = 0, Y = 0) = 0.9$, $P(X = 1, Y = 0) = 0.1$. Under $\mathrm{do}(X = x)$, $Y = x \oplus U$:

	$P(Y = 1 \mid \mathrm{do}(x))$	$a(Y = 1; x)$	$b(Y = 1; x)$
$x = 0$	0.1	0	0.1
$x = 1$	0.9	0	0.9

The true effects lie on the upper bounds.

2.1 Tightness

Theorem 2.3. *For binary X, Y and any $P(x, y)$ there exist SCMs $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ with $P^{(1)}(x, y) = P^{(2)}(x, y) = P(x, y)$ and $P^{(1)}(y \mid \mathrm{do}(x)) = a(y; x)$, $P^{(2)}(y \mid \mathrm{do}(x)) = b(y; x)$.*

Proof. Take any compatible $\mathcal{M}$ ($X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x, u)$) inducing P, and define, with the same $P(U)$,

$$\mathcal{F}^{(1)}:\ Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 0 & \text{otherwise} \end{cases}, \qquad \mathcal{F}^{(2)}:\ Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 1 & \text{otherwise} \end{cases},$$

both with $X \leftarrow f_X(u)$. Without intervention $X = f_X(U)$ always, so $Y = f_Y(X, U)$ in both models and $P^{(i)}(x, y) = \sum_u P(x \mid u)P(y \mid x, u)P(u) = P(y, x)$. Under $\mathrm{do}(X = x)$, U and x need not match:

$$\begin{aligned} P^{(i)}(Y = 1 \mid \mathrm{do}(x)) &= P(Y = 1 \mid x, f_X(U) = x)P(f_X(U) = x) \\ &\quad + P^{(i)}(Y = 1 \mid x, f_X(U) \neq x)P(f_X(U) \neq x) \\ &= P(x, Y = 1) + \mathbb{1}[i = 2](1 - P(x)). \end{aligned}$$

Thus $\mathcal{M}^{(1)}$ attains a and $\mathcal{M}^{(2)}$ attains b (for $Y = 1$; $Y = 0$ is symmetric). □

2.2 Limitations

Natural bounds are tight for the bow graph with binary variables, but not when further structure is available, e.g. an instrument Z. Can the graph be exploited for tighter bounds? Yes: canonical type models and linear programming.

3 Canonical type models and LP bounds

Purpose. Replace the unknown functional forms by a finite set of response types, turning the bounding problem into a linear program.

3.1 Experiments with non-compliance

Setting: treatment assigned Z (randomized, the instrument), treatment realized X, primary outcome Y, and exogenous U confounding X and Y: $Z \to X \to Y$, $X \leftrightarrow Y$. Task: given $P(x, y \mid z)$, bound $P(y \mid \mathrm{do}(x))$.

Theorem 3.1 (Natural bounds in the IV model)**.**

$$a(y; x) = \max_z P(y, x \mid z), \qquad b(y; x) = \min_z \left[P(y, x \mid z) + 1 - P(x \mid z)\right].$$

These follow by applying the natural bounds in each submodel $\mathcal{M}_z$, $P(y, x \mid \mathrm{do}(z)) \leq P(y \mid \mathrm{do}(x, z)) \leq P(y, x \mid \mathrm{do}(z)) + 1 - P(x \mid \mathrm{do}(z))$, using $P(y \mid \mathrm{do}(x, z)) = P(y \mid \mathrm{do}(x))$ (exclusion) and $P(\cdot \mid \mathrm{do}(z)) = P(\cdot \mid z)$ (randomization). They are not tight in IV models.

3.2 Optimization perspective

Let $\mathcal{M}$ be the set of SCMs compatible with the IV graph: $Z \leftarrow f_Z(U_Z)$, $X \leftarrow f_X(Z, U)$, $Y \leftarrow f_Y(X, U)$. The sharp bounds are

$$a(y; x) = \min_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \mathrm{do}(x)), \qquad b(y; x) = \max_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \mathrm{do}(x)).$$

This is hard to solve directly because the forms of $\mathcal{F}$ and $P(\mathbf{U})$ are unspecified.

3.3 U as a selector and response variables

For $X \leftarrow f_X(Z,U)$ and fixed $U=u$, $f_X(\cdot,u)$ is a deterministic function from Z to X: U *selects* which mapping $Z \to X$ is in effect. The domain $\mathfrak{X}_U$ can therefore be partitioned by the induced function. With binary Z, X there are $|\mathfrak{X}_X|^{|\mathfrak{X}_Z|} = 2^2 = 4$ functions:

$$\{u : f_X(0,u)=0,\ f_X(1,u)=0\} \quad (X=0), \qquad \{u : f_X(0,u)=0,\ f_X(1,u)=1\} \quad (X=Z),$$
$$\{u : f_X(0,u)=1,\ f_X(1,u)=0\} \quad (X=1-Z), \qquad \{u : f_X(0,u)=1,\ f_X(1,u)=1\} \quad (X=1).$$

(With ternary Z and binary X there would be 2^3.) A *response variable* indexes this partition:

$R_X=0$: $X=0$ (never-taker) $\quad R_X=1$: $X=Z$ (complier) $\quad R_X=2$: $X=1-Z$ (defier)
$R_X=3$: $X=1$ (always-taker)

Similarly $R_Y \in \{0,1,2,3\}$ for $Y \leftarrow f_Y(X,U)$: never-recover ($Y=0$), helped ($Y=X$), hurt ($Y=1-X$), always-recover ($Y=1$). Together (R_X, R_Y) partition $\mathfrak{X}_U$ into $4\times 4 = 16$ cells with probabilities

$$q_{j,k} = P(R_X=j, R_Y=k), \qquad \sum_{j=0}^{3}\sum_{k=0}^{3} q_{j,k} = 1.$$

Definition 3.2 (Canonical type model)**.** An SCM with $X, Y, Z \in \{0,1\}$ and $\mathbf{U} = \{R_X, R_Y, R_Z\}$, $R_X, R_Y \in \{0,1,2,3\}$, $R_Z \in \{0,1\}$:

$$Z \leftarrow r_Z, \qquad X \leftarrow f_X(z, r_X) = \begin{cases} 0 & r_X = 0 \\ z & r_X = 1 \\ 1-z & r_X = 2 \\ 1 & r_X = 3 \end{cases}, \qquad Y \leftarrow f_Y(x, r_Y) = \begin{cases} 0 & r_Y = 0 \\ x & r_Y = 1 \\ 1-x & r_Y = 2 \\ 1 & r_Y = 3 \end{cases}.$$

Theorem 3.3 (Correspondence; Balke and Pearl 1997)**.** *For any IV model $\mathcal{M}_1$ there is a canonical type model $\mathcal{M}_2$ with $P^{\mathcal{M}_1}(x, y \mid z) = P^{\mathcal{M}_2}(x, y \mid z)$ and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x)) = P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$.*

(Set $q_{j,k}$ to the $P(U)$-mass of the cell of $\mathfrak{X}_U$ inducing the pair of functions (j,k).) Hence, with $\mathcal{M}$ now the canonical models,

$$a(y;x) = \min_{M\in\mathcal{M}} P_M(y \mid \mathrm{do}(x)), \quad b(y;x) = \max_{M\in\mathcal{M}} P_M(y \mid \mathrm{do}(x)) \quad \text{s.t. } P_M(x,y \mid z) = P(x,y \mid z).$$

Summary. U is partitioned into response types by the functions it induces; for the binary IV, R_X (never-taker, complier, defier, always-taker) and R_Y; the joint $P(R_X, R_Y)$ has 16 parameters $q_{j,k}$; every IV model has an equivalent canonical model. Bounding $P(y \mid \mathrm{do}(x))$ reduces to a linear program over $\mathbf{q} = (q_{j,k}) \in \mathbb{R}^{16}$.

3.4 LP formulation

Observational constraints. Let $p_{yx.z} = P(Y=y, X=x \mid Z=z)$. Each is a linear function of $\mathbf{q}$, e.g.

$$\begin{aligned} p_{10.1} = P(Y=1, X=0 \mid Z=1) &= P(\text{never-taker, hurt}) + P(\text{never-taker, always-recover}) \\ &\quad + P(\text{defier, hurt}) + P(\text{defier, always-recover}) \\ &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}. \end{aligned}$$

(With $Z = 1$, $X = 0$ requires $R_X \in \{0, 2\}$; then $Y = 1$ at $X = 0$ requires $R_Y \in \{2, 3\}$.) In full:

$$\begin{aligned}
p_{00.0} &= q_{0,0} + q_{0,1} + q_{1,0} + q_{1,1}, \quad p_{00.1} = q_{0,0} + q_{0,1} + q_{2,0} + q_{2,1}, \\
p_{01.0} &= q_{2,0} + q_{2,2} + q_{3,0} + q_{3,2}, \quad p_{01.1} = q_{1,0} + q_{1,2} + q_{3,0} + q_{3,2}, \\
p_{10.0} &= q_{0,2} + q_{0,3} + q_{1,2} + q_{1,3}, \quad p_{10.1} = q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}, \\
p_{11.0} &= q_{2,1} + q_{2,3} + q_{3,1} + q_{3,3}, \quad p_{11.1} = q_{1,1} + q_{1,3} + q_{3,1} + q_{3,3}.
\end{aligned}$$

Causal effect. Y under $\mathrm{do}(X = 0)$ equals 1 iff $R_Y \in \{\text{hurt}, \text{always-recover}\}$:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) = P(R_Y = 2) + P(R_Y = 3) = \textstyle\sum_j (q_{j,2} + q_{j,3}),$$

$$P(Y = 1 \mid \mathrm{do}(X = 1)) = \textstyle\sum_j (q_{j,1} + q_{j,3}).$$

Optimization. The bound $P(Y = 1 \mid \mathrm{do}(X = 0)) \in [a_{10}, b_{10}]$ solves

$$a_{10} = \min_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3}), \qquad b_{10} = \max_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3})$$

subject to $\sum_{j,k} q_{j,k} = 1$, $q_{j,k} \geq 0$, and the eight observational equalities above.

3.5 Closed-form solution

The LP has a closed-form solution: each bound is a linear combination of the $p_{yx.z}$ whose gap from the objective is provably non-negative.

$$P(Y = 1 \mid \mathrm{do}(X = 0)) \;\geq\; \max \begin{cases} p_{10.0} \\ p_{10.1} \\ p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} \\ p_{01.0} + p_{10.0} - p_{00.1} - p_{01.1} \end{cases}$$

$$P(Y = 1 \mid \mathrm{do}(X = 0)) \;\leq\; \min \begin{cases} 1 - p_{00.0} \\ 1 - p_{00.1} \\ p_{10.0} + p_{11.0} + p_{01.1} + p_{10.1} \\ p_{01.0} + p_{10.0} + p_{10.1} + p_{11.1} \end{cases}$$

The first two lower bounds and upper bounds are the natural bounds; the extra terms make the LP bounds tighter.

Why is $p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}$ a valid lower bound? Substituting,

$$p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} = q_{0,2} + q_{0,3} + q_{1,2} + q_{2,3} - q_{0,0} - q_{0,1} - q_{2,0} - q_{1,1}.$$

Subtracting from the objective,

$$\sum_j (q_{j,2} + q_{j,3}) - (p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}) = q_{2,2} + q_{3,2} + q_{1,3} + q_{3,3} + q_{0,0} + q_{0,1} + q_{2,0} + q_{1,1} \;\geq 0.$$

Every closed-form bound is certified the same way (this is LP duality: the coefficients are a feasible dual solution).

4 Back-door with partially observed covariates

Purpose. Bound an effect when the admissible set is only partly measured but the marginal of the missing part is known.

Suppose $\{W, U\}$ is back-door admissible ($W \to X$, $W \to Y$, $U \to X$, $U \to Y$, W–U related, $X \to Y$), W observed, U not:

$$P(y \mid \mathrm{do}(x)) = \sum_{w,u} P(y \mid x, w, u) P(w, u) = \sum_{w,u} \frac{P(x, y, w, u) P(w, u)}{P(x, w, u)}.$$

Given $P(X, Y, W)$ and $P(U)$ but not $P(X, Y, W, U)$, the naive (natural) bound is $P(x, y) \le P(y \mid \mathrm{do}(x)) \le 1 - P(x, y')$, where $y' = 1 - y$, $x' = 1 - x$ denote complements. Li and Pearl (2022) tighten it:

$$a(y; x) = \min_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}}, \qquad b(y; x) = \max_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}},$$

where $a_{w,u}$, $b_{w,u}$, $c_{w,u}$ stand for the unknown $P(x, y, w, u)$, $P(w, u)$, $P(x, w, u)$, subject to, for all $w \in \mathfrak{X}_W$ and $u \in \mathfrak{X}_U$,

$$\begin{gathered}
\textstyle\sum_u a_{w,u} = P(x, y, w), \qquad \sum_u b_{w,u} = P(w), \qquad \sum_u c_{w,u} = P(x, w), \\
b_{w,u} \ge c_{w,u} \ge a_{w,u} \quad (\text{since } P(x, y, w, u) \le P(x, w, u) \le P(w, u)), \\
\max\{0, P(x, y, w) + P(u) - 1\} \le a_{w,u} \le \min\{P(x, y, w), P(u)\}, \\
\max\{0, P(w) + P(u) - 1\} \le b_{w,u} \le \min\{P(w), P(u)\}, \\
\max\{0, P(x, w) + P(u) - 1\} \le c_{w,u} \le \min\{P(x, w), P(u)\}.
\end{gathered}$$

The last three lines are Fréchet bounds for a joint probability given its two marginals. (The slides remark that $a_{w,u}$ alone is a sufficient parametrization.)

5 Source coverage

The source deck has 57 physical pages (29 numbered slides with build-ups).

Topic	Physical pages
Recall, partial identification problem	1–8
Natural bounds, example, proof, tightness, limitations	9–20
Non-compliance, IV natural bounds, optimization view	21–27
Response variables, canonical type model, correspondence	28–44
LP formulation and closed-form solution	45–55
Partially observed back-door covariates	56–57